Chapter 19 exercises • Windows 7

OS19-E01 • Granted rights and SAME_ACCESS

A's file handle 0x40 points to object F and grants abstract R=1, not W=2. Successful DUPLICATE_SAME_ACCESS gives B handle 0x88; B may also have its own handles.

1. Can A write through 0x40 or B write through 0x88?
2. Must B's 0x40 name the same object? Does A close freeF?
3. Does this subset model describe every different-access DuplicateHandle?

OS19-E02 • Session scope

S1/S2 have interactive users and Local\Ready events. Authorized Global\Ready access is also possible.

1. Which resources are separate/shared?
2. Does equal Local name mean equal object? Does Global bypass DACL?
3. Are these two VMs?

OS19-E03 • Actual ReadFile outcomes

On Windows 7, an unchanged regular file contains ABCDEFGHIJ; no concurrent write or truncation occurs. Open it with FILE_FLAG_OVERLAPPED and request 6 bytes at offsets 0, 6, and 10. Assume the first two reads complete successfully. Each unfinished operation has its own OVERLAPPED, buffer, and event, all valid through terminal completion.

1. Interpret FALSE/ERROR_IO_PENDING. May lpNumberOfBytesRead be NULL?
2. Give final bytes/data and required lifetimes.
3. Does successful CancelIoEx permit immediate free?

OS19-E04 • IRQL and thread priority

A DISPATCH_LEVEL DPC wants a 5 ms I/O wait and potentially faulting paged-pool access while another CPU updates a shared list.

1. Are these operations legal? Define alternative work placement.
2. Does raising local IRQL exclude that CPU?
3. Why does priority 31 not guarantee a hard deadline?

OS19-E05 • Fixed priority queues

P1 and P2 have priority 8, are ready at t=0, and enter the same-level FIFO in P1→P2 order; their CPU bursts are 4 and 3. P3 has priority 10, arrives at t=1, and needs 2. Quantum=2, with no boost/decay; higher priority preempts immediately, and a preempted task retains its remaining slice and returns to the front of its priority queue.

1. Trace intervals/completions.
2. Which premise fails if P1 continues until 2 after P3 arrives at 1?
3. Is q2 universal Windows 7 behavior?

OS19-E06 • Reserve, commit, resident

Reserve 64 KiB at private base 0x100000 and commit two readable/writable 4 KiB pages at 0x101000 and 0x102000. All operations succeed; initially neither page is resident. No other page is loaded or evicted during the trace, and the decommitted page has no other references.

1. Give three quantities before/after touching 0x101123.
2. Decommit first page, then release region. What changes?
3. Does commit immediately zero-write both pagefile pages?

OS19-E07 • Ordered DACL

The token includes Alice and an enabled, non-deny-only Staff SID. Abstract R=1 and W=2 are non-overlapping specific rights after generic mapping. ACE order is allow Alice R, deny Staff W, allow Staff RW. Exclude owner, privileges, restricted SIDs, MIC, and other policy rules.

1. Trace remaining forRW.
2. Move allow Staff R/W before deny. What happens?
3. Are NULL and empty DACL equivalent?

OS19-E08 • NTFS extents

Cluster size is 4096 B, and the volume starts at disk byte 1048576. Extents are (VCN 0,len 4,LCN 100), (4,3,200), (7,2,hole), (9,2,190). The data stream is sparse, logical EOF is 11 clusters, and all requested positions are before EOF. Ignore compression and metadata allocation.

1. Map file byte 20603 to LCN/volume/disk byte.
2. Read VCN7+123 and calculate allocation for 11 logical clusters.
3. What do MFT resident attributes and sequence numbers mean?

OS19-E09 • Single-transaction WAL

Use one isolated metadata transaction; block 200 is reserved exclusively and no other transaction runs concurrently. Old: size 4096, map [100], bitmap[200]=0. New: size 8192, map [100,200], bitmap[200]=1. The complete old/new log and its completeness marker become durable before any home update. Steal and no-force are allowed. No user-data atomicity or durability is assumed.

1. No commit, only home bit became 1. Recover it.
2. Durable commit, all home still old. Recover it.
3. Do checkpoints/metadata recovery guarantee all user data?

OS19-E10 • Snapshots and remote uncertainty

Live X=41/Y=80 creates S; X changes to 42 then 43. Remote append is sent but client times out.

1. Give live/snapshot values and old-backing reclamation condition.
2. Is S necessarily application-consistent and an independent backup?
3. Is retry guaranteed not to duplicate append?

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz et al., Operating System Concepts, 9th ed., Ch. 19. Microsoft: IRQL, VirtualAlloc, AccessCheck. ReadFile, EOF, CancelIoEx, named pipes.

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https://github.com/xinhi6716/computer-science-self-study